TECH ZEXO
STUDY NOTES

Class 10 • Science • Ch-1 | CBSE • NCERT

CHAPTER 01

Chemical Reactions & Equations

ekdum clear notes, mazze ke saath!

NCERT + PYQ + MCQ + Memes

Exam-ready edition



SHIVAM BHAIYA

Namaste dosto! Is chapter ko hum mazze se aur ekdum clear karenge. Chalo shuru!

WHAT'S INSIDE

- Full Notes
- 11 Diagrams
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- NCERT Q&A
- PYQ + MCQ
- Flashcards

MADE WITH ♡ FOR STUDENTS BY TECH ZEXO

TECH ZEXO NOTES

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TECH ZEXO NOTES

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Box colour code *(exam se pehle orange + red boxes padh lo!)*

Definition · Remember · Exam Tip · Common Mistake · Fun Fact · Activity · Why? · Mnemonic

SECTION 01

1 Chemical Reactions

What happens when substances change into new ones

Physical change vs Chemical change

Physical change	Chemical change
No new substance is formed	One or more new substances are formed
Usually reversible (melting ice, dissolving sugar)	Usually irreversible (burning paper, cooking food)
Only physical properties (state, shape) change	Chemical composition changes

Chemical reaction

A process in which one or more substances (**reactants**) change into new substances (**products**) with different properties. Bonds between atoms break and new bonds form; atoms are only rearranged, never created or destroyed. Daily-life examples: milk turning sour, iron rusting, grapes fermenting, food being digested, respiration.

Five signs that a chemical reaction has taken place

Sign	Example
1. Change in state	Burning of a candle (solid wax → gases); Mg ribbon (solid) → white powder
2. Change in colour	Iron nail in blue $CuSO_4$ solution: blue → light green
3. Evolution of a gas	Zinc + dilute H_2SO_4 gives bubbles of hydrogen
4. Change in temperature	Quicklime + water becomes hot
5. Formation of a precipitate	Lead nitrate + potassium iodide gives a yellow solid



SHIVAM BHAIYA

Yaad rakhne ka trick: "State, Colour, Gas, Temp, Ppt", bolo "SCGTP", matlab "Sab Chemistry Gaana Tum Padho" ☺

Burning of magnesium (Activity 1.1)

Magnesium burns with a **dazzling white flame** and changes into a **white powder, magnesium oxide**. $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$. The ribbon is cleaned with sandpaper first to remove the layer of magnesium oxide (and carbonate) on its surface.

Chemical changes around you

Milk turning into curd • cooking food • ripening of fruits • digestion of food • burning LPG • rusting of an iron gate • respiration in every cell of your body.

Why does this happen? Candle?

A burning candle shows **both** changes: wax melting is physical; wax burning to CO_2 and water vapour is chemical.

Common Mistake

Writing **$Mg + O_2 \rightarrow MgO$** in the exam. It is **not balanced**: oxygen is 2 on the left and 1 on the right. Correct: $2Mg + O_2 \rightarrow 2MgO$.

SECTION 02

2 Chemical Equations

The short-hand language of chemistry

Word equation → Skeletal equation

Word equation: Magnesium + Oxygen → Magnesium oxide. Reactants are written on the left, products on the right, joined by an arrow (→) that means "changes into".

Skeletal equation: Mg + O_2 → MgO. It uses formulae but is **not balanced**.

Balanced chemical equation

An equation in which the **number of atoms of each element is equal on both sides**. Balancing follows the **law of conservation of mass**: mass can neither be created nor destroyed in a chemical reaction.

Balancing step by step (NCERT example)

Iron reacts with steam: Fe + H_2O → Fe_3O_4 + H_2

Step	What we do	Equation now
1	Draw boxes around formulae. **Never change a formula**, only put numbers (coefficients) in front.	Fe + H_2O → Fe_3O_4 + H_2
2	Count atoms. Fe: 1 vs 3; H: 2 vs 2; O: 1 vs 4	(unbalanced)
3	Start with the compound having most atoms (Fe_3O_4). Balance O: put 4 before H_2O	Fe + $4H_2O$ → Fe_3O_4 + H_2
4	Now H: 8 on left, so put 4 before H_2	Fe + $4H_2O$ → Fe_3O_4 + $4H_2$
5	Balance Fe: put 3 before Fe	3Fe + $4H_2O$ → Fe_3O_4 + $4H_2$
6	Check every element: Fe 3=3, H 8=8, O 4=4 ✓	Balanced!

HIT & TRIAL METHOD

$$3Fe_{(s)} + 4H_2O_{(g)} \longrightarrow Fe_3O_{4(s)} + 4H_{2(g)}$$

BHAIYA KA NOTE

Balancing ka golden rule: formula kabhi mat chhedo, sirf aage number lagao. H_2O ko H_2O_2 banaya toh woh paani nahi, hydrogen peroxide ban gaya! ☺



Making equations more informative

- **State symbols:** (s) solid, (l) liquid, (g) gas, (aq) aqueous solution (dissolved in water).
- **Conditions** such as temperature, pressure or a catalyst are written above or below the arrow.

$$CO_{(g)} + 2H_{2(g)} \xrightarrow{340\ atm} CH_3OH_{(l)}$$

PHOTOSYNTHESIS

$$6CO_{2(aq)} + 12H_2O_{(l)} \xrightarrow{\text{Sunlight, Chlorophyll}} C_6H_{12}O_{6(aq)} + 6O_{2(aq)} + 6H_2O_{(l)}$$

Exam Tip

In a 'write a balanced equation with state symbols' question, state symbols carry marks. Steam is written $H_2O(g)$, water $H_2O(l)$, a dissolved salt (aq), a precipitate (s).

Practice: balance these (answers on the right)

Skeletal	Balanced
$H_2 + Cl_2 \rightarrow HCl$	$H_2 + Cl_2 \rightarrow 2HCl$
$Na + H_2O \rightarrow NaOH + H_2$	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
$Al + O_2 \rightarrow Al_2O_3$	$4Al + 3O_2 \rightarrow 2Al_2O_3$
$CH_4 + O_2 \rightarrow CO_2 + H_2O$	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Why does this happen? Balance at all?

Atoms are never created or destroyed in a reaction (law of conservation of mass), so every atom on the left must show up on the right.

SECTION 03

3 Types of Chemical Reactions

Combination • Decomposition • Displacement • Double displacement

Combination	Decomposition	Displacement	Double displacement
A + B → AB	AB → A + B	A + BC → AC + B	AB + CD → AD + CB

3.1 Combination reaction

Combination reaction

A reaction in which **two or more reactants combine to form a single product**.

QUICKLIME → SLAKED LIME

$$CaO_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Calcium oxide (quicklime) reacts vigorously with water to form calcium hydroxide (slaked lime), releasing a large amount of heat. A solution of slaked lime is used for **whitewashing**. It reacts slowly with CO_2 in air to form a thin shiny layer of calcium carbonate on the walls (this takes 2–3 days):

$$Ca(OH)_{2(aq)} + CO_{2(g)} \longrightarrow CaCO_{3(s)} + H_2O_{(l)}$$

More examples: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ (burning of coal), $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$ (formation of water).

Fun Fact

Marble's chemical formula is also $CaCO_3$. That's why whitewashed walls look a little shiny after a few days.

3.2 Exothermic & endothermic reactions

Exothermic

Reactions in which **heat is released** along with products. Examples: burning of natural gas, respiration, decomposition of vegetable matter into compost, CaO + water.

Endothermic

Reactions in which **energy is absorbed** (as heat, light or electricity). Examples: most decomposition reactions, photosynthesis, $Ba(OH)_2 + NH_4Cl$ (test tube becomes cold).

BURNING OF NATURAL GAS

$$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(g)}$$

RESPIRATION

$$C_6H_{12}O_{6(aq)} + 6O_{2(aq)} \longrightarrow 6CO_{2(aq)} + 6H_2O_{(l)} + \text{Energy}$$



SHIVAM BHAIYA

Respiration exothermic kyun? Glucose ka oxidation hota hai aur energy release hoti hai. Isi energy se hum padhai karte hain... aur reels bhi dekhte hain 😀

3.3 Decomposition reaction

Decomposition reaction

A reaction in which a **single reactant breaks down into two or more simpler products**. It needs energy, so decomposition reactions are generally **endothermic**.

Type	Energy as	Example
Thermal decomposition	Heat	$FeSO_4$, $CaCO_3$, $Pb(NO_3)_2$
Electrolytic decomposition (electrolysis)	Electricity	Water → H_2 + O_2
Photolytic decomposition (photolysis)	Light	AgCl, AgBr

(a) Thermal decomposition

GREEN → REDDISH-BROWN

$$2FeSO_4(s) \xrightarrow{\text{Heat}} Fe_2O_3(s) + SO_2(g) + SO_3(g)$$

Green ferrous sulphate crystals ($FeSO_4 \cdot 7H_2O$) first lose water of crystallisation; then $FeSO_4$ decomposes to ferric oxide with the characteristic **smell of burning sulphur** (SO_2, SO_3).

LIMESTONE → QUICKLIME

$$CaCO_3(s) \xrightarrow{\text{Heat}} CaO(s) + CO_2(g)$$

Used to make quicklime, which is used in the manufacture of cement.

BROWN FUMES OF NO_2

$$2Pb(NO_3)_2(s) \xrightarrow{\text{Heat}} 2PbO(s) + 4NO_2(g) + O_2(g)$$

(b) Electrolytic decomposition

$$2H_2O(l) \xrightarrow{\text{Electric current}} 2H_2(g) + O_2(g)$$

Hydrogen collects at the cathode (−) and oxygen at the anode (+). The volume of hydrogen is **double** that of oxygen because water has 2 H atoms for every O atom (2 : 1 by volume).

(c) Photolytic decomposition

WHITE → GREY

$$2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$$

$$2AgBr(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Br_2(g)$$

These reactions are used in **black and white photography**. Silver salts are kept in dark-coloured bottles.

Common Mistake

Thinking all decomposition is by heat. Ask: **what energy breaks it?** Heat = thermal, electricity = electrolytic, light = photolytic.

3.4 Displacement reaction

Displacement reaction

A reaction in which a **more reactive element displaces a less reactive element** from its compound (in solution).

BLUE → LIGHT GREEN

$$Fe_{(s)} + CuSO_{4(aq)} \longrightarrow FeSO_{4(aq)} + Cu_{(s)}$$

The iron nail becomes brownish (copper deposits on it) and the blue colour of copper sulphate fades to light green (ferrous sulphate). Iron is more reactive than copper.

$$Zn_{(s)} + CuSO_{4(aq)} \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$$

$$Pb_{(s)} + CuCl_{2(aq)} \longrightarrow PbCl_{2(aq)} + Cu_{(s)}$$

Reactivity order you need here (most → least)

K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au

BHAIYA KA NOTE

Simple funda: bada (zyada reactive) chhote ko bahar nikaal deta hai. Copper, iron ko $FeSO_4$ se nahi nikaal sakta, isliye $Cu + FeSO_4 \rightarrow$ no reaction!



3.5 Double displacement reaction

Double displacement reaction

A reaction in which **two compounds exchange their ions** to form two new compounds.

WHITE PPT

$$Na_2SO_{4(aq)} + BaCl_{2(aq)} \longrightarrow BaSO_{4(s)} + 2NaCl_{(aq)}$$

Precipitation reaction

Any reaction that produces an **insoluble solid (precipitate)**. Barium sulphate (white) and lead iodide (yellow) are precipitates.

YELLOW PPT

$$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \longrightarrow PbI_{2(s)} + 2KNO_{3(aq)}$$

Another example (neutralisation, also double displacement): $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

Exam Tip

Displacement = **one** element + **one** compound. Double displacement = **two** compounds swapping partners. Count the 'singles' on the reactant side and you'll never confuse them.

SECTION 04

4 Oxidation & Reduction

Redox: the give-and-take of oxygen and hydrogen

Oxidation

Gain of oxygen or **loss of hydrogen** by a substance.

Reduction

Loss of oxygen or **gain of hydrogen** by a substance.

Redox reaction

A reaction in which **one reactant gets oxidised while the other gets reduced**. Oxidation and reduction always happen together.

When copper powder is heated in air it turns black (Activity 1.11): $2Cu + O_2 \rightarrow (Heat)\ 2CuO$. Passing hydrogen over hot black CuO gives back brown copper:

$$CuO + H_2 \xrightarrow{Heat} Cu + H_2O$$

Reaction	Oxidised (gains O / loses H)	Reduced (loses O / gains H)
$CuO + H_2 \rightarrow Cu + H_2O$	$H_2 \rightarrow H_2O$	$CuO \rightarrow Cu$
$ZnO + C \rightarrow Zn + CO$	$C \rightarrow CO$	$ZnO \rightarrow Zn$
$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$	$HCl \rightarrow Cl_2$ (loses H)	$MnO_2 \rightarrow MnCl_2$ (loses O)

Oxidising agent: the substance that gives oxygen (or removes hydrogen); it gets reduced itself (CuO, ZnO, MnO_2 above). **Reducing agent:** the substance that removes oxygen (or gives hydrogen); it gets oxidised itself (H_2, C, HCl above).



SHIVAM BHAIYA

Exam tip! Question mein agar "substance oxidised" pucha hai toh reactant ka naam likho (jaise H_2), product (H_2O) ka nahi. Yahin pe sabse zyada marks katte hain!

Common Mistake

In $2PbO + C \rightarrow 2Pb + CO_2$, students write 'lead is reduced' or 'CO_2 is oxidised'. Correct: **PbO is reduced** (it loses O) and **C is oxidised** (it gains O). That's NCERT Exercise Q1.

SECTION 05

5 Corrosion & Rancidity

Oxidation in daily life

Corrosion

When a metal is attacked by substances around it such as **moisture, air (oxygen) and acids**, it slowly gets eaten up. This is corrosion.

- **Iron** → reddish-brown **rust** (hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$).
- **Silver** → black coating (silver sulphide).
- **Copper** → green coating (basic copper carbonate).

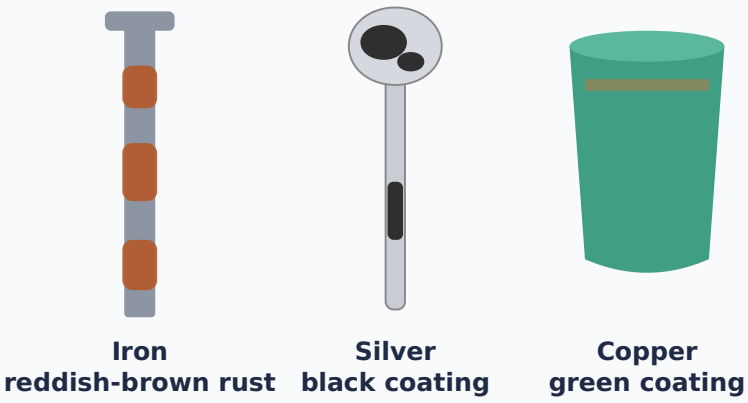


Fig. 1.11 Corrosion of common metals

Corrosion damages car bodies, bridges, iron railings, ships and all metal objects, especially iron. A lot of money is spent every year to replace damaged iron.

Prevention: painting, oiling, greasing, galvanisation (zinc coating), chrome plating, alloying.

Rancidity

When fats and oils are **oxidised**, they become rancid: their **smell and taste change**.

Prevention: adding **antioxidants**; storing food in **air-tight containers**; keeping in the refrigerator; **flushing packets with nitrogen** (chips packets), which removes oxygen.

BHAIYA KA NOTE

Chips ka packet aadha "khaali" nahi hai bhai, usme nitrogen bhari hai taaki chips rancid na ho. Ab dukaan wale se ladna band! 😄



BONUS PAGE

B1 Reaction Types at a Glance

Ek table, poora chapter

Type & pattern	NCERT example	Real-life example
Combination A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$	Whitewashing walls; burning coal $C + O_2 \rightarrow CO_2$
Thermal decomposition AB → A + B (heat)	$CaCO_3 \xrightarrow{\text{Heat}} CaO + CO_2$	Making quicklime for cement
Electrolytic decomposition AB → A + B (electricity)	$2H_2O \rightarrow 2H_2 + O_2$	Getting reactive metals (Na, Al) by electrolysis
Photolytic decomposition AB → A + B (light)	$2AgBr \xrightarrow{\text{light}} 2Ag + Br_2$	Black-and-white photography
Displacement A + BC → AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	Thermite welding of rail tracks $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$
Double displacement (precipitation) AB + CD → AD + CB	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$	Antacid neutralising extra stomach acid (neutralisation is also double displacement)
Exothermic Heat given out	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Burning LPG / natural gas; respiration keeps us warm
Endothermic Heat taken in	$Ba(OH)_2 + NH_4Cl$: test tube turns cold	Photosynthesis; cooking food
Redox Oxidation + reduction together	$CuO + H_2 \rightarrow Cu + H_2O$	Getting metals from ores; rusting; burning fuels

One reaction, many labels

A single reaction can belong to more than one type. Burning of methane is **combustion**, **exothermic** and **redox** (CH_4 is oxidised). $CaO + H_2O \rightarrow Ca(OH)_2$ is both **combination** and **exothermic**. Respiration is **exothermic** and a kind of **oxidation**.

BHAIYA KA NOTE

Exam mein "identify the type" aaye toh pehle pattern dekho: kitne reactants, kitne products, kaun kisko hata raha hai. Phir energy check karo.



B2

BONUS PAGE

Colour & Precipitate Chart

Observation questions ka shortcut

Ferrous sulphate heated (Act. 1.6)

Green crystals → **Reddish-brown Fe_2O_3 + burning-sulphur smell**

Iron nail in $CuSO_4$ (Act. 1.9)

Blue solution → **Light green $FeSO_4$; brown Cu coats nail**

Lead nitrate + potassium iodide

Two colourless solutions → **Yellow precipitate of PbI_2**

Sodium sulphate + barium chloride

Two colourless solutions → **White precipitate of $BaSO_4$**

Silver chloride in sunlight

White → **Grey (silver metal forms)**

Copper powder heated (Act. 1.11)

Brown copper → **Black copper(II) oxide, CuO**

Lead nitrate heated (Act. 1.7)

White crystals → **Brown fumes of NO_2**

Magnesium ribbon burnt (Act. 1.1)

Silvery ribbon → **Dazzling white flame, white MgO powder**

Iron left in moist air

Grey iron → **Reddish-brown rust**

Silver articles in air

Shiny silver → **Black coating (silver sulphide)**

Copper vessels in moist air

Brown copper → **Green coating (basic copper carbonate)**

CO_2 through lime water

Clear lime water → **Milky ($CaCO_3$ forms)**



SHIVAM BHAIYA

Colour wale 1-mark questions pakke hain. Peela = PbI_2, Safed = $BaSO_4$, aur blue se green = iron ne copper hata diya!

BONUS PAGE

B3 Memory Tricks & Mnemonics

Ratto mat, trick lagao

Reactivity series (most → least reactive)

K	Na	Ca	Mg	Al	Zn	Fe	Pb	H	Cu	Hg	Ag	Au
P	*S*	*C*	*M*	*A*	*Z*	*I*	*L*	*H*	*C*	*M*	*S*	*G*

"Please Stop Calling Me A Zebra, I Like Her Calling Me Smart Goat"

Use: a metal **higher** in the list displaces a metal **lower** in the list from its salt solution. Fe beats Cu, so Fe + $CuSO_4$ works; Cu + $FeSO_4$ gives no reaction.

Oxidation vs Reduction

"O aaya, H gaya = Oxidation"

Oxidation: gain of O / loss of H

Reduction: loss of O / gain of H

Bonus (Class 11): **OIL RIG** = Oxidation Is Loss, Reduction Is Gain (of electrons).

Agent ulta kaam karta hai

"Oxidising agent khud reduce hota hai"

$CuO + H_2 \rightarrow Cu + H_2O$: CuO gives O, so CuO is the **oxidising agent** and is itself **reduced**. H_2 is the **reducing agent** and is **oxidised**.

Decomposition: HEL → TEP

Heat → **T**hermal

Electricity → **E**lectrolytic

Light → **P**hotolytic

"Help Ek Ladke ki, TEP lagao"

Electrolysis of water: PANIC

Positive **A**node, **N**egative **I**s **C**athode.

H_2 collects at the cathode (−), O_2 at the anode (+). H_2 volume is **double**: 2 : 1, because H_2O has 2 H for every O.

Corrosion colours

"Lal Loha, Hara Tamba, Kaali Chandi"

Iron → reddish-brown rust • Copper → green coat • Silver → black coat

Stop rancidity: A-N-A-R (anar!)

Antioxidants • **N**itrogen flushing • **A**ir-tight containers • **R**efrigeration

Precipitate colours

"Peela Pb, Safed Ba"

PbI_2 = yellow ppt • $BaSO_4$ = white ppt

Signs of a reaction: SCGTP

State • **C**olour • **G**as • **T**emperature • **P**recipitate

"Sab Chemistry Gaana Tum Padho"

BONUS PAGE

B4 Why Does This Happen?

Reason samjho, ratta khatam

Why is the Mg ribbon rubbed with sandpaper first?

Its surface has a coating of magnesium oxide (and carbonate) that stops it from burning easily. Sandpaper exposes fresh Mg.

Why does the beaker get hot when quicklime meets water?

$CaO + H_2O \rightarrow Ca(OH)_2$ is a combination reaction that **releases heat**: exothermic.

Why is the H_2 volume double the O_2 volume?

Water is H_2O: two H for every one O. So $2H_2O \rightarrow 2H_2 + O_2$ gives gases in a **2 : 1** ratio.

Why is dilute H_2SO_4 added in electrolysis of water?

Pure water conducts electricity very poorly. A few drops of acid make it conduct.

Why are silver salts stored in dark bottles?

Light decomposes silver chloride/bromide (photolytic decomposition), turning them grey.

Why does blue $CuSO_4$ turn light green with an iron nail?

Iron is more reactive than copper, so it displaces Cu. The $FeSO_4$ formed is light green.

Why does whitewash give a shiny finish?

$Ca(OH)_2$ reacts slowly with CO_2 in air to form a thin layer of $CaCO_3$ after 2–3 days.

Why is respiration called exothermic?

Glucose combines with oxygen in our cells and **energy is released**.

Why are chips packets flushed with nitrogen?

Nitrogen is unreactive. It pushes out oxygen, so the oils in chips don't get oxidised (no rancidity).

Why does iron not rust in dry air?

Rusting needs **both** oxygen and moisture. Remove either one and rusting stops.

BHAIYA KA NOTE

"Give reason" wale 2-markers yahin se aate hain. Reason ke saath equation likh do toh full marks!



TECH ZEXO NOTES

Meme Corner

Hasao, aur yaad karao: har meme ek concept

#1 BALANCING EQUATIONS IN THE EXAM



✗ **H_2O ko H_2O_2 bana diya. Oxygen balance ho gaya na? Topper move.**



✓ **Formula ko haath bhi nahi lagaya, bas aage number:**
$2H_2 + O_2 \rightarrow 2H_2O$

Seekh: Balance sirf coefficients (aage ke number) se. Subscript badla toh substance hi badal gaya: H_2O_2 = hydrogen peroxide!

#2 POV: IRON NAIL NE $CuSO_4$ KE GROUP MEIN ENTRY MAARI

Copper
Bhai main yahan pehle se hoon, sulphate mera hai 😠

Iron
Reactivity series dekh le. Main tujhse upar hoon 👀

Copper
nail pe brown layer bann ke baith gaya ☹

Solution
Blue se light green. Breakup official ♡

Seekh: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$: zyada reactive metal kam reactive metal ko hata deta hai (displacement).

#3 EXPECTATION vs REALITY: CHEMISTRY PRACTICAL

EXPECTATION



Movie wala dhamaka, aag, slow-motion walk out

REALITY



Test tube bas thoda garam hua. Sir: "exothermic hai, observation likho."

Seekh: Heat nikle = **exothermic**. $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$ + heat. Garam beaker hi asli dhamaka hai.

#4 NOBODY: / CHEMISTRY MA'AM:

Nobody:
Absolutely nobody:
Ma'am: *"Beta state symbols kahan hain? (s), (l), (g), (aq)... warna half mark gaya."*



Seekh: (s) solid, (l) liquid, (g) gas, (aq) paani mein ghula. Steam = $H_2O(g)$, paani = $H_2O(l)$, precipitate = (s).

#5 ME: / ALSO ME:



Me: Chips ko rancid hone se bachane ke liye air-tight dabba use karunga.



Also me: Packet 3 din khula chhod ke: "ye chips ajeeb smell kyun kar rahe hain?"

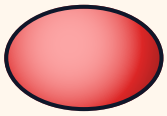
Seekh: Rancidity = fats/oils ka oxidation, smell aur taste kharab. Rokne ke liye: antioxidants, nitrogen flushing, air-tight dabba, fridge.

#6 DISPLACEMENT REACTION BE LIKE (distracted boyfriend edition)

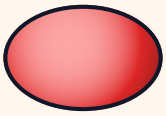


Seekh: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$: sulphate zyada reactive zinc ke saath chala jaata hai, copper akela reh jaata hai.

#7 THE TWO BUTTONS: AgCl DHOOP MEIN RAKHA HAI



Thermal decomposition



Photolytic decomposition



board exam mein pasina pasina

Seekh: Dhoop = light = **photolytic**. $2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$ (white se grey). Heat = thermal, bijli = electrolytic.

#8 LEVELS OF UNDERSTANDING RUST

Rust = loha ganda ho gaya
Rust = iron + paani
Rust = iron + oxygen + moisture, dheere dheere (corrosion)
Rust = hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$. Rokna hai? Paint, oil, galvanising!

Seekh: Board answer mein last wala level likho: naam + formula + conditions + prevention.

#9 CLASS 10 CHEM CH-1 STARTER PACK

- Blue $CuSO_4$ ka light green ho jaana
- Yellow ppt = PbI_2
- Electrolysis: $H_2 : O_2 = 2 : 1$
- Chips packet mein nitrogen
- Mg ki dazzling white flame
- H_2 ka "pop" sound test

Seekh: Ye 6 observations sabse zyada pooche jaate hain. Ek baar inka colour/sign yaad, toh 1-markers free.

#10 FAREWELL PARTY MEIN EVERYONE vs ME



Sab log: naach rahe hain



Me in the corner: "Inko pata bhi nahi ki saans lena (respiration) ek exothermic reaction hai."

Seekh: $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l)$ + energy. Respiration se hi body ko energy milti hai.

Meme challenge

Ab tumhari baari! Is chapter ke kisi bhi reaction pe apna meme banao aur study group mein share karo. Ideas: **Iron vs Copper rivalry**, **Ma'am vs state symbols**, **AgCl ka Goa trip** (white se grey), **chips packet ki nitrogen**.



SHIVAM BHAIYA

Meme yaad reh gaya toh concept bhi yaad reh gaya. Doston ko bhejo, saath mein padho!

SECTION 07

7 NCERT Activities

Aim • Materials • Procedure • Observation • Conclusion

Activity numbers follow the NCERT textbook edition most guides use (1.1–1.11). Some newer reprints renumber a few: match the activity by its name.

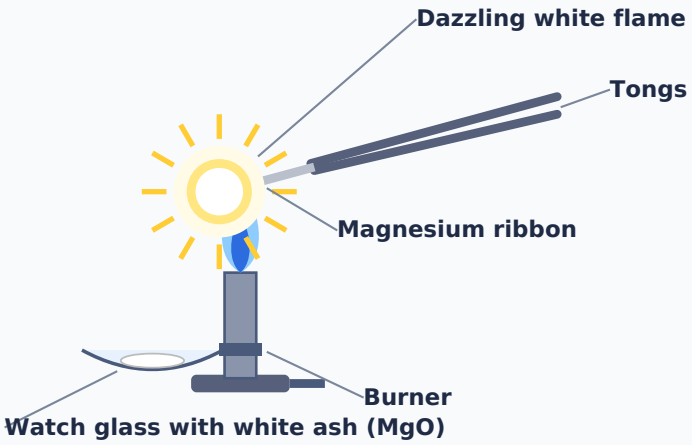


Fig. 1.1 Burning of a magnesium ribbon in air and collection of magnesium oxide (white ash) in a watch glass

Activity 1.1: Burning a magnesium ribbon

Aim: To observe the burning of magnesium in air.

Materials: Magnesium ribbon (2 cm), sandpaper, tongs, burner, watch glass.

Procedure:

1. Clean the ribbon with sandpaper.
2. Hold it with tongs and burn it over a burner, keeping it away from your eyes.
3. Collect the ash in a watch glass.

Observation: Mg burns with a dazzling white flame and leaves a white powder.

Conclusion: Magnesium combines with oxygen to form magnesium oxide: $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$. (Combination, exothermic)

Activity 1.2: Lead nitrate + potassium iodide

Aim: To observe a precipitate forming.

Materials: Lead nitrate solution, potassium iodide solution, test tube.

Procedure:

1. Take lead nitrate solution in a test tube.
2. Add potassium iodide solution to it.

Observation: A yellow precipitate forms.

Conclusion: Lead iodide (yellow ppt) is formed: $Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)} + 2KNO_{3(aq)}$. (Double displacement, precipitation)

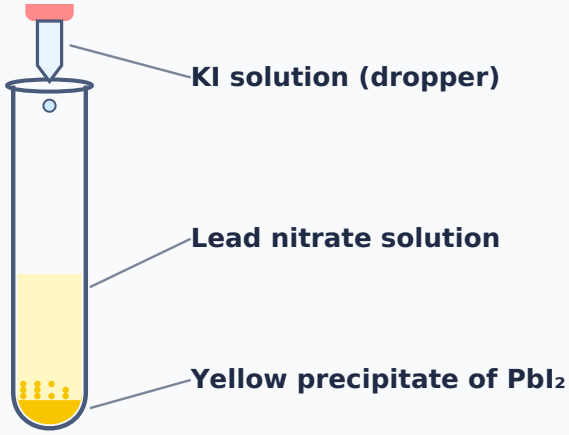


Fig. 1.2 Formation of a yellow precipitate of lead iodide

Activity 1.3: Zinc + dilute acid

Aim: To observe the evolution of a gas and the change in temperature.

Materials: Zinc granules, dilute HCl or H_2SO_4, conical flask or test tube.

Procedure:

1. Put zinc granules in a flask or test tube.
2. Add dilute acid and touch the flask.

Observation: Bubbles of gas form around the granules; the flask becomes warm.

Conclusion: Hydrogen gas is evolved: $Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)}$. The gas burns with a pop sound. (Displacement, exothermic)

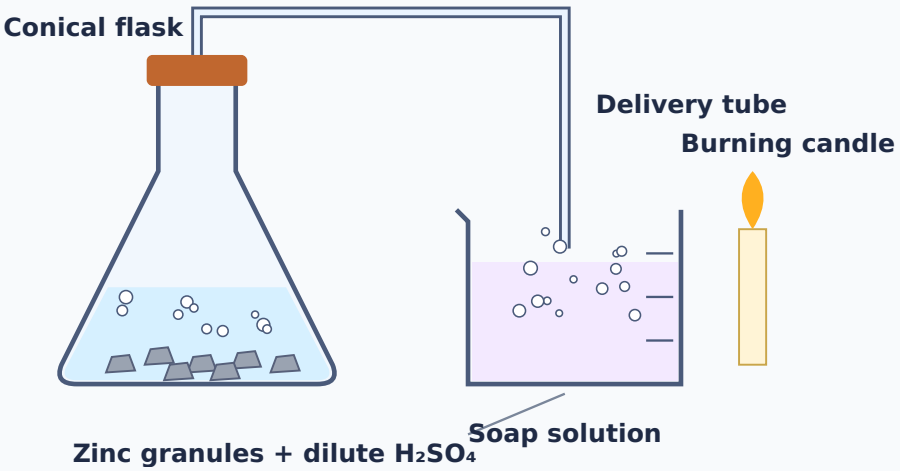


Fig. 1.3 Hydrogen gas formed by the action of dilute sulphuric acid on zinc; soap bubbles filled with it burn with a pop

Activity 1.4: Quicklime + water

Aim: To study a combination reaction.

Materials: Calcium oxide (quicklime), water, beaker.

Procedure:

1. Take a small amount of CaO in a beaker.
2. Slowly add water and touch the beaker.

Observation: A hissing sound; the beaker becomes very hot.

Conclusion: Slaked lime is formed with release of heat: $CaO_{(s)} + H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)}$ + heat. (Combination, exothermic)

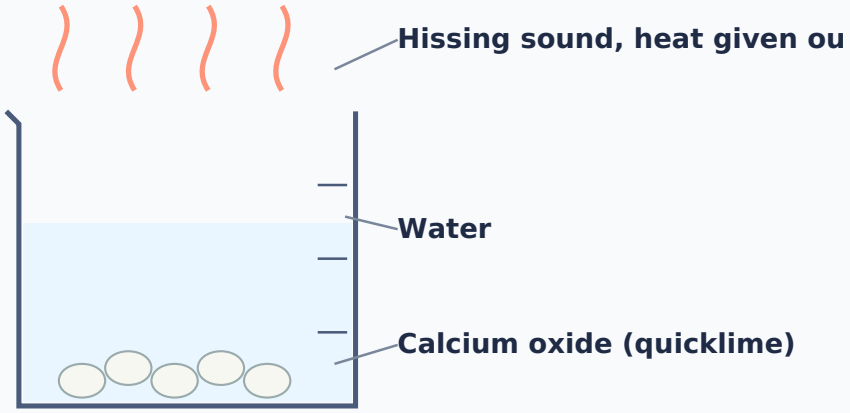


Fig. 1.4 Quicklime reacts vigorously with water: slaked lime forms and heat is released

Activity 1.5: Heating ferrous sulphate crystals

Aim: To study thermal decomposition.

Materials: $FeSO_4$ crystals (about 2 g), dry boiling tube, test-tube holder, burner.

Procedure:

1. Take crystals in a dry boiling tube.
2. Heat over a burner and observe the colour; smell the gas carefully (waft).

Observation: Green crystals lose water, then turn into a reddish-brown solid; a smell of burning sulphur.

Conclusion: $FeSO_4$ decomposes: $2FeSO_{4(s)} \xrightarrow{\text{(Heat)}} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$. (Thermal decomposition)

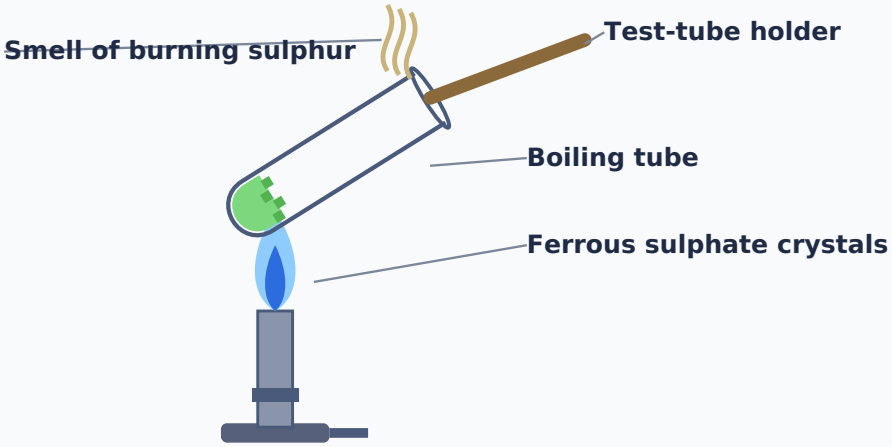


Fig. 1.5 Heating ferrous sulphate crystals: green crystals turn into a reddish-brown solid with a sulphur smell

Activity 1.6: Heating lead nitrate

Aim: To study thermal decomposition with a gas.

Materials: Lead nitrate powder (2 g), boiling tube, burner, holder.

Procedure:

1. Heat lead nitrate in a boiling tube over a flame.

Observation: Brown fumes are given off and a yellowish solid remains.

Conclusion: Nitrogen dioxide (brown) is released:
$2Pb(NO_3)_2(s) \xrightarrow{\text{Heat}} 2PbO(s) + 4NO_2(g) + O_2(g)$. (Thermal decomposition)

Activity 1.7: Electrolysis of water

Aim: To decompose water using electricity.

Materials: Plastic mug, two graphite electrodes, 6 V battery, two test tubes, water with a few drops of dilute H_2SO_4.

Procedure:

1. Fix two carbon electrodes in the mug and fill with water + dilute acid.
2. Invert two water-filled test tubes over the electrodes.
3. Switch on the current for some time.

Observation: Gas bubbles at both electrodes; the gas at the cathode is double in volume. H_2 burns with a pop; O_2 relights a glowing splinter.

Conclusion: Water decomposes: $2H_2O(l) \xrightarrow{\text{Electricity}} 2H_2(g) + O_2(g)$. (Electrolytic decomposition)

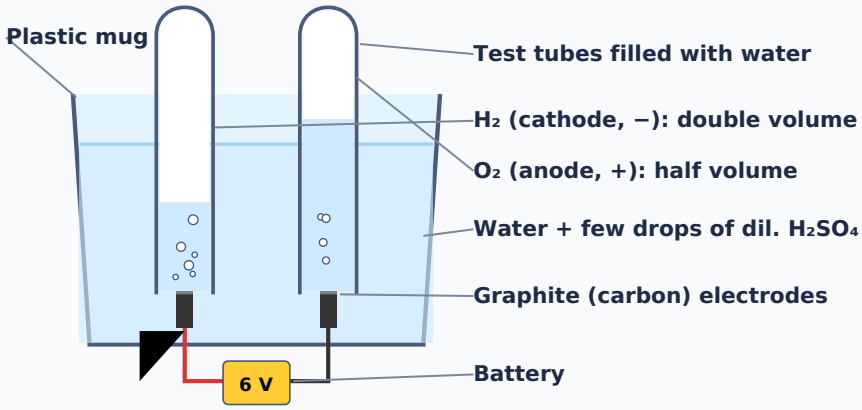


Fig. 1.6 Electrolysis of water: hydrogen collects at the cathode, oxygen at the anode (volume ratio 2 : 1)

Activity 1.8: Silver chloride in sunlight

Aim: To study photolytic decomposition.

Materials: Silver chloride (about 2 g), china dish, sunlight.

Procedure:

1. Take AgCl in a china dish.
2. Keep it in sunlight for some time.

Observation: White AgCl turns grey.

Conclusion: Silver chloride decomposes in light: $2AgCl_{(s)} \xrightarrow{\text{Sunlight}} 2Ag_{(s)} + Cl_{2(g)}$. (Photolytic decomposition)

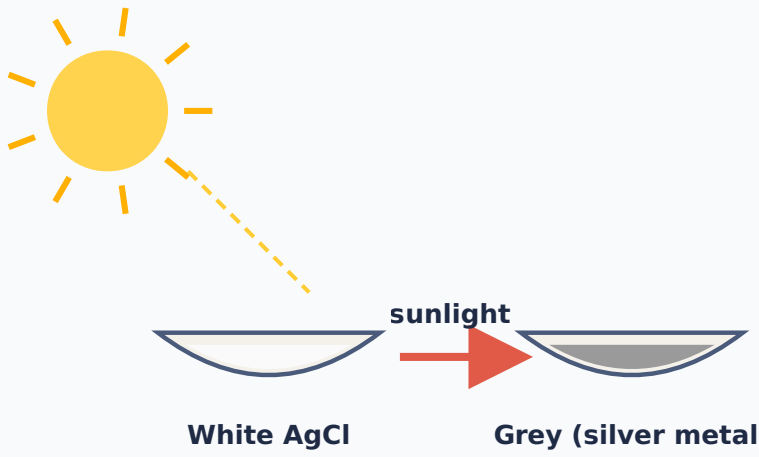


Fig. 1.7 Silver chloride turns grey in sunlight as it decomposes into silver and chlorine

Activity 1.9: Iron nails in copper sulphate

Aim: To study a displacement reaction.

Materials: 3 iron nails, sandpaper, two test tubes (A, B), copper sulphate solution.

Procedure:

1. Clean the nails with sandpaper.
2. Dip two nails in $CuSO_4$ solution in tube A for about 20 min; keep tube B as control.
3. Compare nail and solution colours.

Observation: Nails become brownish; blue solution turns light green.

Conclusion: Iron displaces copper: $Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$. (Displacement)

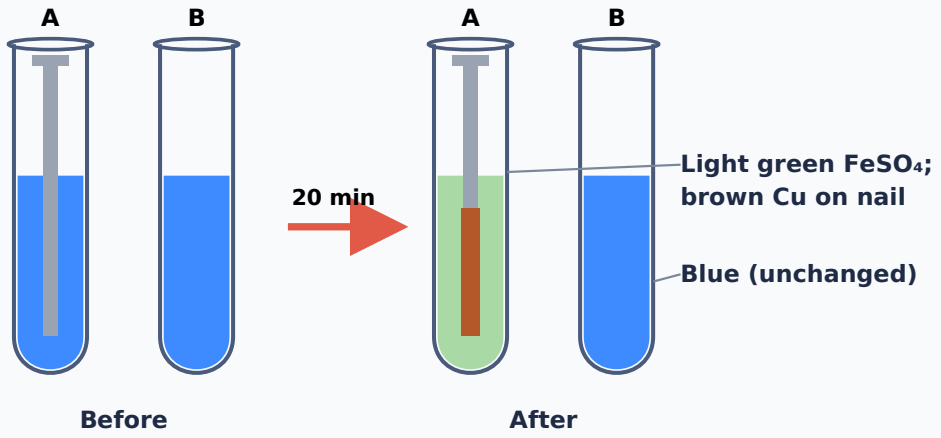


Fig. 1.8 Iron nails dipped in copper sulphate solution (A) compared with a control tube (B)

Activity 1.10: Sodium sulphate + barium chloride

Aim: To study a double displacement reaction.

Materials: Sodium sulphate solution (3 mL), barium chloride solution (3 mL), test tubes.

Procedure:

1. Take Na_2SO_4 solution in one test tube and $BaCl_2$ solution in another.
2. Mix them.

Observation: A white insoluble substance (precipitate) forms.

Conclusion: Barium sulphate is formed: $Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)} + 2NaCl_{(aq)}$. (Double displacement, precipitation)

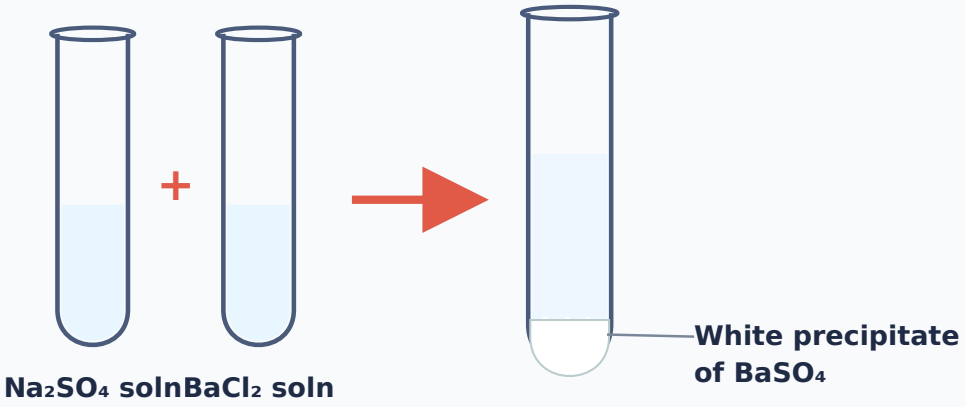


Fig. 1.9 Mixing sodium sulphate and barium chloride solutions gives a white precipitate of barium sulphate

Activity 1.11: Heating copper powder

Aim: To study oxidation.

Materials: Copper powder (1 g), china dish, tripod, wire gauze, burner.

Procedure:

1. Take copper powder in a china dish.
2. Heat it on a wire gauze.

Observation: The brown surface of copper powder turns black.

Conclusion: Copper is oxidised to copper(II) oxide: $2Cu + O_2 \xrightarrow{\text{(Heat)}} 2CuO$. If H_2 is passed over hot CuO, it turns brown again (reduction).

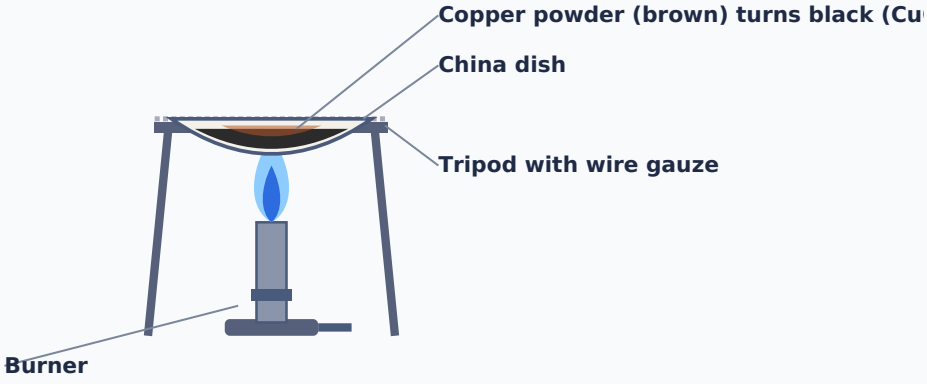


Fig. 1.10 On heating in air, brown copper powder gets coated with black copper(II) oxide



SHIVAM BHAIYA

Har activity ka observation yaad rakho: rang (colour), gas, garmi (heat) ya ppt. Board exam mein yahi pucha jaata hai!

SECTION 08

8 NCERT In-text Questions

All in-text questions with answers

Q1. Why should a magnesium ribbon be cleaned before burning in air?

ANSWER

Magnesium reacts with air to form a layer of magnesium oxide (and carbonate) on its surface. This layer stops the ribbon from burning properly, so it is removed with sandpaper.

Q2. Write the balanced equation for: (i) Hydrogen + Chlorine → Hydrogen chloride (ii) Barium chloride + Aluminium sulphate → Barium sulphate + Aluminium chloride (iii) Sodium + Water → Sodium hydroxide + Hydrogen

ANSWER

$$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$$

$$3BaCl_2(aq) + Al_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2AlCl_3(aq)$$

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$$

Q3. Write a balanced equation with state symbols: (i) Barium chloride and sodium sulphate solutions react to give insoluble barium sulphate and sodium chloride solution. (ii) Sodium hydroxide solution reacts with hydrochloric acid solution to give sodium chloride solution and water.

ANSWER

$$BaCl_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(s) + 2NaCl(aq)$$

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

Q4. A solution of a substance 'X' is used for whitewashing. (i) Name X and write its formula. (ii) Write the reaction of X with water.

ANSWER

(i) X is calcium oxide (quicklime), CaO. (ii)

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq) + Heat$$

Q5. Why is the amount of gas collected in one test tube in Activity 1.7 double that in the other? Name this gas.

ANSWER

Water (H_2O) has two hydrogen atoms for every oxygen atom, so on electrolysis hydrogen and oxygen form in a 2 : 1 volume ratio. The gas with double volume is **hydrogen**.

Q6. Why does the colour of copper sulphate solution change when an iron nail is dipped in it?

ANSWER

Iron is more reactive than copper, so it displaces copper from copper sulphate. Blue $CuSO_4$ changes to light green $FeSO_4$: **$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$**

Q7. Give an example of a double displacement reaction other than the one in Activity 1.10.

ANSWER

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$ or $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

Q8. Identify the substances oxidised and reduced: (i) $4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$ (ii) $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$

ANSWER

(i) Na is oxidised (gains oxygen); O_2 is reduced. (ii) H_2 is oxidised (gains oxygen); CuO is reduced (loses oxygen).

SECTION 09

9 NCERT Exercise Solutions

Exercise Q1–Q20

BHAIYA KA NOTE

NCERT exercise = board exam ka syllabus ka dil ♡. Inhe 2 baar zaroor likh ke practice karo.



Q1. Which statements about 2PbO(s) + C(s) → 2Pb(s) + CO_2(g) are incorrect? (a) Lead is getting reduced (b) Carbon dioxide is getting oxidised (c) Carbon is getting oxidised (d) Lead oxide is getting reduced. Options: (i) (a) and (b) (ii) (a) and (c) (iii) (a), (b) and (c) (iv) all

ANSWER

(i) (a) and (b). It is lead oxide (not lead) that is reduced, and carbon (not CO_2) that is oxidised.

Q2. Fe_2O_3 + 2Al → Al_2O_3 + 2Fe is an example of: (a) combination (b) double displacement (c) decomposition (d) displacement

ANSWER

(d) Displacement: Al displaces Fe from its oxide.

Q3. What happens when dilute hydrochloric acid is added to iron filings? (a) Hydrogen gas and iron chloride are produced (b) Chlorine gas and iron hydroxide (c) No reaction (d) Iron salt and water

ANSWER

(a) $Fe_{(s)} + 2HCl_{(aq)} \rightarrow FeCl_{2(aq)} + H_{2(g)}$

Q4. What is a balanced chemical equation? Why should chemical equations be balanced?

ANSWER

An equation with equal numbers of atoms of each element on both sides. It must be balanced to obey the law of conservation of mass: atoms are neither created nor destroyed in a reaction.

Q5. Translate and balance: (a) Hydrogen + nitrogen → ammonia (b) Hydrogen sulphide burns in air → water + sulphur dioxide (c) Barium chloride + aluminium sulphate → aluminium chloride + barium sulphate (d) Potassium + water → potassium hydroxide + hydrogen

ANSWER

$$3H_{2(g)} + N_{2(g)} \longrightarrow 2NH_{3(g)}$$

$$2H_2S_{(g)} + 3O_{2(g)} \longrightarrow 2H_2O_{(l)} + 2SO_{2(g)}$$

$$3BaCl_{2(aq)} + Al_2(SO_4)_{3(aq)} \longrightarrow 2AlCl_{3(aq)} + 3BaSO_{4(s)}$$

$$2K_{(s)} + 2H_2O_{(l)} \longrightarrow 2KOH_{(aq)} + H_{2(g)}$$

Q6. Balance: (a) $HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$ (b) $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$ (c) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$ (d) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$

ANSWER

$$2HNO_3 + Ca(OH)_2 \longrightarrow Ca(NO_3)_2 + 2H_2O$$

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$

(c) is already balanced.

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$$

Q7. Write balanced equations: (a) Calcium hydroxide + carbon dioxide → calcium carbonate + water (b) Zinc + silver nitrate → zinc nitrate + silver (c) Aluminium + copper chloride → aluminium chloride + copper (d) Barium chloride + potassium sulphate → barium sulphate + potassium chloride

ANSWER

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$Zn + 2AgNO_3 \longrightarrow Zn(NO_3)_2 + 2Ag$$

$$2Al + 3CuCl_2 \longrightarrow 2AlCl_3 + 3Cu$$

$$BaCl_2 + K_2SO_4 \longrightarrow BaSO_4 + 2KCl$$

Q8. Write the balanced equation and identify the type: (a) $KBr(aq) + BaI_2(aq) \rightarrow KI(aq) + BaBr_2(s)$ (b) $ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)$ (c) $H_2(g) + Cl_2(g) \rightarrow HCl(g)$ (d) $Mg(s) + HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

ANSWER

DOUBLE DISPLACEMENT

$$2KBr_{(aq)} + BaI_{2(aq)} \longrightarrow 2KI_{(aq)} + BaBr_{2(s)}$$

DECOMPOSITION

$$ZnCO_{3(s)} \xrightarrow{\text{Heat}} ZnO_{(s)} + CO_{2(g)}$$

COMBINATION

$$H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$$

DISPLACEMENT

$$Mg_{(s)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + H_{2(g)}$$

Q9. What do you mean by exothermic and endothermic reactions? Give examples.

ANSWER
Exothermic: heat is released, e.g. burning of natural gas $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, respiration. Endothermic: energy is absorbed, e.g. $CaCO_3 \rightarrow$ (Heat) $CaO + CO_2$, photosynthesis, electrolysis of water.

Q10. Why is respiration considered an exothermic reaction?

ANSWER
During respiration glucose combines with oxygen in our cells and releases energy: $C_6H_{12}O_6(aq) + 6O_2(aq) \rightarrow 6CO_2(aq) + 6H_2O(l) +$ **Energy**. Since energy is released, it is exothermic.

Q11. Why are decomposition reactions called the opposite of combination reactions? Write equations.

ANSWER
In combination, two or more reactants form one product; in decomposition, one reactant breaks into two or more products. $CaO + CO_2 \rightarrow CaCO_3$ (combination) and $CaCO_3 \rightarrow$ (Heat) $CaO + CO_2$ (decomposition). Also $2H_2 + O_2 \rightarrow 2H_2O$ vs $2H_2O \rightarrow$ (Electricity) $2H_2 + O_2$

Q12. Write one equation each for decomposition where energy is supplied as heat, light and electricity.

ANSWER
Heat: $CaCO_3(s) \rightarrow$ (Heat) $CaO(s) + CO_2(g)$
Light: $2AgCl(s) \rightarrow$ (Sunlight) $2Ag(s) + Cl_2(g)$
Electricity: $2H_2O(l) \rightarrow$ (Electricity) $2H_2(g) + O_2(g)$

Q13. Difference between displacement and double displacement reactions, with equations.

ANSWER
Displacement: a more reactive element displaces a less reactive one from its compound, $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. Double displacement: two compounds exchange ions, $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Q14. In the refining of silver, silver is recovered from silver nitrate solution by displacement with copper. Write the reaction.

ANSWER

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

Q15. What is a precipitation reaction? Explain with examples.

ANSWER
A reaction that forms an insoluble solid (precipitate). $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$ (white ppt); $Pb(NO_3)_2 + 2KI \rightarrow PbI_2\downarrow + 2KNO_3$ (yellow ppt).

Q16. Explain in terms of gain or loss of oxygen, with two examples each: (a) Oxidation (b) Reduction

ANSWER
(a) Oxidation = gain of oxygen: $2Cu + O_2 \rightarrow 2CuO$; $C + O_2 \rightarrow CO_2$. (b) Reduction = loss of oxygen: $CuO + H_2 \rightarrow Cu + H_2O$ (CuO reduced); $ZnO + C \rightarrow Zn + CO$ (ZnO reduced).

Q17. A shiny brown element 'X' turns black on heating in air. Name X and the black compound.

ANSWER
X is **copper (Cu)**; the black compound is **copper(II) oxide (CuO)**: $2Cu + O_2 \rightarrow$ (Heat) $2CuO$

Q18. Why do we apply paint on iron articles?

ANSWER
Paint keeps air and moisture away from the iron surface, which prevents rusting (corrosion).

Q19. Oil and fat containing food items are flushed with nitrogen. Why?

ANSWER
Nitrogen is unreactive and replaces oxygen in the packet, so the oils and fats are not oxidised and do not become rancid.

Q20. Explain with an example: (a) Corrosion (b) Rancidity

ANSWER
(a) Corrosion: a metal is slowly eaten up by moisture, air, acids, e.g. rusting of iron (reddish-brown), black coating on silver. (b) Rancidity: oxidation of fats and oils changes their smell and taste, e.g. butter or fried snacks kept open for long smell bad.

SECTION 10

10 CBSE PYQ-style Questions

Repeated board themes

These follow the questions CBSE asks again and again from this chapter. We have **not** put years on them, because we couldn't verify each year; treat them as **CBSE-style** practice.

Q1. Write the chemical equation for the reaction taking place when quicklime reacts with water. Name the type of reaction and state one use of the product. **2 marks**

ANSWER

$$CaO_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)} + \text{Heat}$$

Combination and exothermic reaction. Slaked lime is used for whitewashing.

Q2. Name the reducing agent in: $ZnO + C \rightarrow Zn + CO$ **1 mark**

ANSWER

Carbon (C) is the reducing agent; it removes oxygen from ZnO and is itself oxidised to CO.

Q3. When ferrous sulphate crystals are heated, what colour change is seen? Write the equation and name the type of reaction. **3 marks**

ANSWER

Green crystals turn into a reddish-brown solid (Fe_2O_3) with a smell of burning sulphur.

$2FeSO_{4(s)} \rightarrow (\text{Heat})\ Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$. Thermal decomposition.

Q4. Why are silver salts stored in dark-coloured bottles? Write the equation involved. **2 marks**

ANSWER

Light decomposes silver salts: $2AgCl_{(s)} \rightarrow (\text{Sunlight})\ 2Ag_{(s)} + Cl_{2(g)}$. Dark bottles keep light out.

Q5. What is observed when a solution of potassium iodide is added to lead nitrate solution? Name the type of reaction and write the balanced equation. **3 marks**

ANSWER

A yellow precipitate of lead iodide forms. Double displacement (precipitation):

$Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \rightarrow PbI_{2(s)} + 2KNO_{3(aq)}$

Q6. Can copper displace iron from iron sulphate solution? Give a reason. **1 mark**

ANSWER

No. Copper is less reactive than iron, so it cannot displace iron.

Q7. Identify the type of each reaction: (a) $2H_2O \rightarrow 2H_2 + O_2$ (b) $NaOH + HCl \rightarrow NaCl + H_2O$ (c) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ **3 marks**

ANSWER

(a) Electrolytic decomposition (b) Double displacement (neutralisation) (c) Displacement

Q8. What is rancidity? Write two ways to prevent it. **2 marks**

ANSWER

Oxidation of fats and oils in food that changes their smell and taste. Prevent by adding antioxidants; storing in air-tight containers; flushing packets with nitrogen; refrigeration.

Q9. Balance the equation $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$ and identify the substance oxidised and the substance reduced. **3 marks**

ANSWER

$$Fe_2O_3(s) + 2Al(s) \longrightarrow Al_2O_3(s) + 2Fe(l)$$

Al is oxidised (gains oxygen); Fe_2O_3 is reduced (loses oxygen). This highly exothermic reaction (thermit reaction) is used to join broken railway tracks.

Q10. Differentiate between exothermic and endothermic reactions with one example each. **2 marks**

ANSWER

Exothermic: heat is released, e.g. $C(s) + O_2(g) \rightarrow CO_2(g)$ + heat. Endothermic: heat is absorbed, e.g. $CaCO_3(s) \rightarrow$ (Heat) $CaO(s) + CO_2(g)$



SHIVAM BHAIYA

Exam tip! PYQ ke themes baar baar repeat hote hain: quicklime + water, $FeSO_4$ heating, AgCl in sunlight, $Pb(NO_3)_2 + KI$, redox identify. Ye 5 pakke kar lo!

SECTION 11

11 MCQs

1 mark each

1. The reaction $CaO + H_2O \rightarrow Ca(OH)_2$ is:
(a) decomposition (b) combination (c) displacement (d) double displacement

2. Which is a physical change?
(a) Rusting of iron (b) Burning of paper (c) Melting of wax (d) Souring of milk

3. Gas evolved when zinc reacts with dilute H_2SO_4:
(a) O_2 (b) H_2 (c) SO_2 (d) CO_2

4. Colour of ferric oxide formed on heating $FeSO_4$:
(a) green (b) white (c) reddish-brown (d) black

5. In electrolysis of water, the ratio of volumes of H_2 to O_2 is:
(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 4 : 1

6. Silver chloride turns grey in sunlight due to:
(a) Ag (b) Cl_2 (c) AgO (d) Ag_2O

7. The white precipitate in $Na_2SO_4 + BaCl_2$ is:
(a) NaCl (b) $BaCl_2$ (c) $BaSO_4$ (d) Na_2SO_4

8. In $CuO + H_2 \rightarrow Cu + H_2O$, the substance oxidised is:
(a) CuO (b) H_2 (c) Cu (d) H_2O

9. Which metal can displace copper from $CuSO_4$ solution?
(a) Silver (b) Gold (c) Zinc (d) Mercury

10. Chips packets are flushed with:
(a) oxygen (b) nitrogen (c) hydrogen (d) chlorine

11. Which is an endothermic process?
(a) Respiration (b) Burning of coal (c) Decomposition of $CaCO_3$ (d) CaO + water

12. Rust is chemically:
(a) FeO (b) Fe_3O_4 (c) $Fe_2O_3 \cdot xH_2O$ (d) $FeCO_3$

13. Brown fumes on heating lead nitrate are of:
(a) O_2 (b) NO_2 (c) N_2O (d) PbO

14. Balanced form of $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ has the coefficient of H_2O as:
(a) 2 (b) 3 (c) 4 (d) 1

15. Which is a redox reaction?
(a) $NaOH + HCl \rightarrow NaCl + H_2O$ (b) $ZnO + C \rightarrow Zn + CO$ (c) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ (d) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

Answer Key

1 – (b) 2 – (c) 3 – (b) 4 – (c) 5 – (b) 6 – (a) 7 – (c) 8 – (b) 9 – (c) 10 – (b)
11 – (c) 12 – (c) 13 – (b) 14 – (c) 15 – (b)

SECTION 12

12 Assertion–Reason

Choose the correct option

Options

(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. (e) Both A and R are false (used in some papers).

1. **Assertion (A): Mg ribbon is cleaned with sandpaper before burning.**
Reason (R): It removes the protective layer of magnesium oxide from the surface.

ANSWER (a) Both true; R explains A.

2. **A: The amount of H_2 formed in electrolysis of water is double that of O_2.**
R: Water contains hydrogen and oxygen in the ratio 2 : 1 by number of atoms.

ANSWER (a) Both true; R explains A.

3. **A: Copper can displace iron from $FeSO_4$ solution.**
R: Iron is more reactive than copper.

ANSWER (d) A is false (Cu is less reactive); R is true.

4. **A: Decomposition of calcium carbonate is an endothermic reaction.**
R: Calcium oxide is used in the manufacture of cement.

ANSWER (b) Both true, but R does not explain A (heat is absorbed to break $CaCO_3$).

5. **A: In $CuO + H_2 \rightarrow Cu + H_2O$, CuO is reduced.**
R: CuO loses oxygen.

ANSWER (a) Both true; R explains A.

6. **A: Chips packets are flushed with oxygen.**
R: Oxygen prevents rancidity.

ANSWER (e) Both A and R are false: packets are flushed with nitrogen; oxygen causes rancidity.

BHAIYA KA NOTE

MCQ mein 12+ sahi aaye? Topper vibes! A–R ka funda: pehle A aur R alag alag sach hain ya nahi, phir dekho R, A ko explain karta hai ya nahi.



SECTION 13

13 Case-based Questions

Read the passage, then answer

Case 1: Rahul's school lab

Rahul dipped a clean iron nail in blue copper sulphate solution. After 20 minutes the nail looked brownish and the solution became light green. In another test tube he mixed sodium sulphate and barium chloride solutions and saw a white solid.

Q1. Name the brown coating on the nail.

ANSWER
Copper metal (Cu).

Q2. Write the balanced equation for the first reaction and name its type.

ANSWER
$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$, displacement.

Q3. Name the white solid and the type of the second reaction.

ANSWER
Barium sulphate ($BaSO_4$); double displacement (precipitation).

Q4. Would the colour change if a silver wire were used instead of the iron nail? Why?

ANSWER
No. Silver is less reactive than copper, so it cannot displace copper.

Case 2: Old snacks, old railing

Priya noticed the namkeen left open for a week smelled and tasted bad, while a sealed packet was fine. The iron railing on her balcony had reddish-brown flakes, but the freshly painted gate did not.

Q5. Name the process that spoilt the namkeen. What causes it?

ANSWER
Rancidity, caused by oxidation of the fats and oils.

Q6. Why was the sealed packet fine?

ANSWER
It was air-tight and flushed with nitrogen, so oxygen could not oxidise the oils.

Q7. Name the reddish-brown flakes and the two things needed for them to form.

ANSWER
Rust (hydrated iron(III) oxide); air (oxygen) and moisture (water).

Q8. How did paint protect the gate?

ANSWER
It formed a layer that kept air and moisture away from the iron.

SECTION 14

14 VSA • SA • LA

1, 2–3 and 5 mark questions

Very Short Answer (1 mark)

Q1. Define a chemical equation.

ANSWER
A symbolic representation of a chemical reaction using formulae of reactants and products.

Q2. What does the symbol (aq) mean?

ANSWER
Aqueous: the substance is dissolved in water.

Q3. Name the gas that burns with a pop sound.

ANSWER
Hydrogen.

Q4. Name one reaction used in black and white photography.

ANSWER
$2AgBr \xrightarrow{\text{(Light)}} 2Ag + Br_2$

Q5. Is burning of natural gas exothermic or endothermic?

ANSWER
Exothermic.

Short Answer (2–3 marks)

Q1. Balance and add state symbols: $Al + O_2 \rightarrow Al_2O_3$

ANSWER

$$4Al_{(s)} + 3O_{2(g)} \longrightarrow 2Al_2O_{3(s)}$$

Q2. Give one example each of a combination reaction that is exothermic and a decomposition reaction that is endothermic.

ANSWER
Exothermic combination: $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} + \text{Heat}$. Endothermic decomposition: $CaCO_{3(s)} \xrightarrow{\text{(Heat)}} CaO_{(s)} + CO_{2(g)}$

Q3. What is the role of a few drops of dilute sulphuric acid in electrolysis of water?

ANSWER
Pure water is a poor conductor of electricity; the acid makes it conduct, so electrolysis can take place.

Q4. A student heats lead nitrate in a test tube. What does he observe? Write the equation.

ANSWER

Brown fumes of NO_2 and a yellowish residue of PbO.

$2Pb(NO_3)_2(s) \rightarrow (Heat)\ 2PbO(s) + 4NO_2(g) + O_2(g)$

Long Answer (5 marks)

Q1. (a) Define combination, decomposition, displacement and double displacement reactions with one balanced equation each. (b) Which of these is generally endothermic and why?

ANSWER

(a) Combination: $CaO + H_2O \rightarrow Ca(OH)_2$; Decomposition: $CaCO_3 \rightarrow (Heat)\ CaO + CO_2$; Displacement: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$; Double displacement: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$. (b) Decomposition, because energy (heat, light or electricity) is needed to break the reactant into simpler products.

Q2. (a) What is a redox reaction? Identify the substance oxidised, reduced, the oxidising agent and the reducing agent in $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$. (b) Explain corrosion and two methods to prevent it.

ANSWER

(a) A reaction in which one substance is oxidised and another is reduced. HCl is oxidised (loses H) and is the reducing agent; MnO_2 is reduced (loses O) and is the oxidising agent. (b) Corrosion: slow eating up of metals by air, moisture and acids, e.g. rusting of iron. Prevention: painting, oiling/ greasing, galvanising.



SHIVAM BHAIYA

Long answer mein definition + balanced equation + example, teeno likhna. Equation balanced nahi hai toh marks kat sakte hain!

BONUS PAGE

B5 Common Mistakes

Ye galtiyan = marks ka nuksaan

⊗ Wrong	✓ Right	Why
~~$Mg + O_2 \rightarrow MgO$~~	✓ **$2Mg + O_2 \rightarrow 2MgO$**	Count atoms of every element on both sides.
~~Balancing by changing H_2O to H_2O_2~~	✓ **$2H_2O$ (coefficient only)**	Changing a formula changes the substance.
~~In $CuO + H_2 \rightarrow Cu + H_2O$: "$H_2O$ is oxidised"~~	✓ **H_2 is oxidised**	Name the **reactant** that changes.
~~Rust is Fe_2O_3~~	✓ **Rust is hydrated iron(III) oxide, $Fe_2O_3 \cdot xH_2O$**	Water is part of rust.
~~PbI_2 is a white precipitate~~	✓ **PbI_2 is a yellow precipitate**	$BaSO_4$ is the white one.
~~Electrolysis gives O_2 double of H_2~~	✓ **$H_2 : O_2 = 2 : 1$; H_2 at cathode**	From the formula H_2O.
~~$Cu + FeSO_4 \rightarrow CuSO_4 + Fe$~~	✓ **No reaction**	Cu is less reactive than Fe.
~~$2FeSO_4 \rightarrow Fe_2O_3 + SO_2$~~	✓ **$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$**	Both oxides of sulphur form.
~~Oxidising agent gets oxidised~~	✓ **Oxidising agent gets reduced**	It gives away its oxygen.
~~Rancidity is caused by germs~~	✓ **Rancidity is oxidation of fats and oils**	That's why nitrogen/ antioxidants help.
~~Forgetting state symbols when asked~~	✓ **Write (s), (l), (g), (aq)**	They carry marks.
~~Decomposition reactions release heat~~	✓ **Most decomposition reactions are endothermic**	Energy is needed to break the compound.
~~$Ca(OH)_2$ is quicklime~~	✓ **CaO is quicklime; $Ca(OH)_2$ is slaked lime**	Quicklime + water → slaked lime.
~~Mg burns with a blue flame~~	✓ **Mg burns with a dazzling white flame**	Ash is white MgO powder.
~~$FeSO_4$ solution is blue~~	✓ **$FeSO_4$ solution is light green**	$CuSO_4$ is the blue one.



SHIVAM BHAIYA

Ruko! Ye page exam se ek din pehle zaroor padhna. Har saal yahi galtiyan sabse zyada marks katwati hain.

B6

BONUS PAGE

Quick-recall Flashcards

Fold karo, khud se poochho

Cover the answer with your hand (or fold the page), say the answer aloud, then check. Dashed lines = cut-out cards.

Q1. What is a balanced equation?

Same number of atoms of each element on both sides.

Q2. Law behind balancing?

Law of conservation of mass.

Q3. $CaO + H_2O$ type?

Combination + exothermic.

Q4. Gas at the cathode in electrolysis?

Hydrogen (double volume).

Q5. Colour of PbI_2 ppt?

Yellow.

Q6. AgCl in sunlight?

Turns grey: $2AgCl \rightarrow 2Ag + Cl_2$ (photolytic).

Q7. Fe + $CuSO_4$ colour change?

Blue → light green.

Q8. Brown fumes on heating $Pb(NO_3)_2$?

Nitrogen dioxide, NO_2.

Q9. Oxidation (Class 10)?

Gain of oxygen / loss of hydrogen.

Q10. Reducing agent in ZnO + C → Zn + CO?

Carbon (C).

Q11. Rust formula?

$Fe_2O_3 \cdot xH_2O$

Q12. Why galvanise iron?

Zinc coating stops rusting.

Q13. Rancidity?

Oxidation of fats/oils: bad smell & taste.

Q14. Respiration is ...?

Exothermic.

TECH ZEXO NOTES

One-page Revision

Night-before-exam sheet

Type	Pattern	Must-know equation
Combination	A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$ + heat
Thermal decomposition	AB → A + B (heat)	$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$
Electrolytic decomposition	electricity	$2H_2O \rightarrow 2H_2 + O_2$
Photolytic decomposition	light	$2AgCl \rightarrow 2Ag + Cl_2$
Displacement	A + BC → AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double displacement	AB + CD → AD + CB	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
Redox	gain/loss of O or H	$CuO + H_2 \rightarrow Cu + H_2O$

Colours to remember

PbI_2 yellow ppt • $BaSO_4$ white ppt • Fe_2O_3 reddish-brown • $CuSO_4$ blue → $FeSO_4$ light green • CuO black • AgCl white → grey • NO_2 brown fumes • Mg flame dazzling white

Top 5 exam points

1. Balance every equation.
2. Add state symbols when asked.
3. Oxidised/reduced: name the reactant.
4. $H_2 : O_2 = 2 : 1$ in electrolysis.
5. Nitrogen flushing prevents rancidity.



Chapter complete! Tumne kar dikhaya. Ab ise apne doston ke saath share karo aur aise aur notes ke liye Tech Zexo follow karo. All the best for boards!